

A Cognitively-Rich Human-AI Interaction Pattern for Deeper Vocabulary Understanding

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Learning new words — understanding nuances, recognizing them in unfamiliar contexts, and producing them naturally — is a cognitively-rich activity [1]. It requires higher-order thinking: abstracting a flexible meaning from specific uses, and re-concretizing it into new ones. Yet prevalent study practices commonly treat vocabulary as lookup — definition in, recognition out — fostering familiarity but not the generative, context-sensitive mastery of real understanding.

Cognitive-science research points to interactions that build such mastery. In Chi and Wylie's ICAP framework, learning deepens as learners move from passive to active to constructive to interactive — producing novel examples, explaining in one's own words, co-constructing knowledge through dialogue [2]. The *Involvement Load Hypothesis* predicts that effortful tasks inducing higher *need*, *search*, and *evaluation* yield better vocabulary learning gains [3].

LLMs make it feasible to scaffold such engagement at scale, with growing evidence that ChatGPT-based interventions improve learning outcomes across domains [4]. LLM chatbots outperform conventional drills in language learning broadly [5] and vocabulary specifically [6], and context personalization sustains motivation [7]. Beyond delivery, how the AI is framed matters: dialogic scaffolding deepens engagement in adjacent tasks such as critical paper reading [8], and framing AI output as questions rather than answers improves reasoning [9]. These findings support the view that AI should be *extraheric* — fostering higher-order thinking through strategies like *questioning* and *scaffolding* rather than just supplying direct answers [10]. Related works in vocabulary and note-taking settings are emerging [11, 12].

This proposal designs and evaluates a human-AI interaction pattern grounded in these principles. The interface will be assessed holistically against learners' current self-directed practice—where LLMs are already common—through two research questions: (RQ1) *learning gains* in contextual recognition and productive use; (RQ2) *learner experience*, specifically (a) cognitive engagement — useful (germane) vs. unnecessary (extraneous) effort — and (b) perceived effectiveness, efficiency, enjoyability, and adoption willingness.

1. Proposed System

For each target word in its most frequent sense, the interface guides the learner through five steps:

1. **Contextual input.** Three AI-generated flashcards — sample sentences with visual aid — that use the word in different contexts are shown one after another.
2. **Elicited meaning.** The AI asks, “What do you think this word means?” The learner types an informed guess; the AI responds with concise targeted feedback.
3. **Productive generation.** The AI asks the learner to describe a scenario in which the word could be used and to write a sample sentence based on their understanding. The AI gives feedback on contextual fit.
4. **Continued dialogue.** The learner may continue the conversation based on their specific needs.
5. **Integrated assessment.** A short quiz closes the block: per word, three contextual multiple-choice items (“does the word make sense in this sentence?” options phrased as 'True because...' / 'False because...', requiring reasoning rather than pattern-matching) and two productive items (“write two sentences using this word in a new fitting scenario”).

2. Methodology*

Participants. ~24 adult non-native English speakers (CEFR B1+, verified via a brief free online placement test, e.g., engxam.com/english-level-test or Strommen's) who self-report wanting to improve their academic English vocabulary.

Design. Within-subjects, blocked crossover. Each participant completes two conditions:

- **App:** learning via the proposed interface (~15–20 min per block)
- **Baseline:** participants are presented with a standard stimulus (definition + sample sentence) and then study using their usual method (including LLMs if preferred) for a matched time, capturing current real-world practice.

The same quiz (proposed system's step 5) is administered at the end of the matched study time in both conditions.

Procedure. Eight sessions over ~one week:

- **Session 0 (screening, ~20 min).** Participants mark ~150 words from a high-frequency academic word list (academicvocabulary.info) as "know / can use" vs. "don't know or can't use." Per participant, 30 eligible words are randomly split 15/15 across conditions, with best-effort balancing for frequency band and POS.

- **Sessions 1–6 (one block per day, 5 new words per block, ~15–20 min).** Participants are randomly assigned to one of two groups. Group A uses the app for Days 1–3 and their usual self-study method for Days 4–6; Group B does the reverse. Flashcards and scripted prompts, quiz-items are pre-piloted and verified against a pre-specified rubric; learner-reactive AI responses are spot-checked post-session for appropriateness.
- **Session 7 (~15 min).** Semi-structured interview and end-of-study questionnaire.

*~32 participants will be recruited to account for attrition. Participants will be compensated for their time. All procedures will be reviewed by an institutional ethics board; informed consent will be obtained, with withdrawal permitted at any time. The analysis plan will be pre-registered before data collection. Participants missing >1 session are excluded from primary analysis, robustness checks include their available data.

Measures. *RQ1 (learning gains):* three contextual multiple-choice items per word and two productive items per word (Step 5 sentence-writing, distinct from Step 3; scored by two blind raters on a form/meaning/context rubric; weighted κ reported). *RQ2 (learner experience):* per block, Paas mental-effort item, two Likert items for germane and extraneous engagement, confidence rating, 3-item Likert battery (perceived effectiveness, efficiency, enjoyability), and tool-use checklist; at study end, summative versions of the per-block experience ratings for each condition, adoption-preference battery, per-component (the five steps) helpfulness/enjoyability ratings, and a semi-structured interview probing all RQs, frustrations and desired changes. All interactions (full conversation) in the app are logged.

Analysis. Learning gains (RQ1) and self-report measures (RQ2) are compared across the two conditions using mixed effect models that account for each participant completing both conditions and for words varying in difficulty, with self-reported LLM use included as a covariate. We check whether results depend on which condition participants did first (to rule out novelty effects) and whether the app still outperforms the subset of baseline blocks where participants used an LLM. Adoption preferences between conditions are analyzed using paired tests appropriate to each item's scale. Interviews are thematically coded by two coders, with agreement reported. AI responses are verified post-hoc against a pre-specified rubric; primary analyses include all trials, with flagged trials excluded as a robustness check. A power analysis will be conducted during piloting to confirm sensitivity to medium effects.

Limitations. The baseline permits unrestricted tool use (including LLMs), so RQ1 measures against real-world practice rather than isolated mechanisms — covariate and subset analyses unpack this. Only immediate comprehension is measured; retention and deeper dimensions (collocation, register) are out of scope. Modest sample size and session count limit generalizability, the controlled setting cannot fully simulate natural study habits, day-to-day variability (mood, motivation) could influence individual ratings; and novelty effects could inflate app performance despite crossover. LLM response variability means feedback differs across participants.

3. Contribution

This study contributes an empirically-evaluated human-AI interaction pattern that engages learners in cognitively-rich vocabulary study, toward deeper word understanding. More broadly, it contributes to the literature on *extraheric* AI [10] — human-AI interactions that deepen rather than replace cognitive engagement.

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